

---

LECTURE NOTES

# Lecture 14: Linear Constrained Optimization

---

SDU

Simon Holm  
April - 2026



DEPARTMENT OF MATHEMATICS  
AND COMPUTER SCIENCE

## Contents

|       |                                                        |   |
|-------|--------------------------------------------------------|---|
| 1     | Linear Programming .....                               | 3 |
| 1.1   | Problem Formulation .....                              | 3 |
| 1.1.1 | Numerical example .....                                | 3 |
| 1.1.2 | General example .....                                  | 4 |
| 1.2   | Conversion to linear program .....                     | 4 |
| 1.3   | Forms .....                                            | 4 |
| 1.3.1 | Standard form .....                                    | 4 |
| 1.3.2 | Equality form .....                                    | 5 |
| 1.3.3 | Quick example .....                                    | 5 |
| 2     | Simplex Algorithm .....                                | 5 |
| 2.1   | Problem Formulation .....                              | 6 |
| 2.2   | Info abt the simplex algorithm .....                   | 6 |
| 2.3   | The Simplex Algorithm .....                            | 7 |
| 2.3.1 | The problem setup .....                                | 7 |
| 3     | Duality .....                                          | 8 |
| 4     | Primal-Dual Hybrid Gradient (PDHG) Method for LP ..... | 8 |

# 1 Linear Programming

## 1.1 Problem Formulation

If an optimization problem has a linear objective and constraints, it is called a **linear programming problem (linear program, LP)**

- The general form is

$$\begin{aligned} \min_x \quad & c^T x \\ \text{s.t.} \quad & Ax \leq b \\ & Dx \geq e \\ & Fx = G \\ & x, c \in \mathbb{R}^n, \\ & A \in \mathbb{R}^{m \times n}, b \in \mathbb{R}^m \\ & D \in \mathbb{R}^{p \times n}, e \in \mathbb{R}^p \\ & F \in \mathbb{R}^{q \times n}, g \in \mathbb{R}^q \end{aligned}$$

### 1.1.1 Numerical example

$$\begin{aligned} & \underset{x_1, x_2, x_3}{\text{minimize}} \quad 2x_1 - 3x_2 + 7x_3 \\ & \text{subject to} \quad 2x_1 + 3x_2 - 8x_3 \leq 5 \\ & \quad \quad \quad 4x_1 + x_2 + 3x_3 \leq 9 \\ & \quad \quad \quad x_1 - 5x_2 - 3x_3 \geq -4 \\ & \quad \quad \quad x_1 + x_2 + 2x_3 = 1 \end{aligned}$$

$$\begin{bmatrix} 2 & 3 & -8 \\ 4 & 1 & 3 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \leq \begin{bmatrix} 5 \\ 9 \end{bmatrix}$$

Where

$$x_1, x_2, x_3 \in \mathbb{R}$$

### 1.1.2 General example

#### Example

Given a set of items  $I$ , each item with a price  $p_i$  and a value  $v_i$  for  $i$  in  $I$ , select the subset of items that maximizes the total value collected subject to a total expense that does not exceed a given budget  $B$ .

$$\begin{aligned} \max \quad & \sum_{i \in I} p_i x_i \\ \text{s.t.} \quad & \sum_{i \in I} v_i x_i \leq B \\ & x_i \in \{0, 1\}, \quad \text{for all } i \text{ in } I \end{aligned}$$

Binary programming works better. So if you had to choose an amount of each item, you should make each item (if you have 10 phones =  $\{x_1, \dots, x_{10}\}$ ) a parameter.

## 1.2 Conversion to linear program

Many problems can be converted into linear programs that have the same solution.

#### Example

$$\text{minimize } L_1 = \|Ax - b\|_1$$

$$\begin{aligned} \min \quad & \mathbf{1}^T \mathbf{s} \\ \text{s.t.} \quad & Ax - b \leq \mathbf{s} \\ & -(Ax - b) \leq \mathbf{s} \end{aligned}$$

Sum of absolute errors  
Robust regression: Ignores outliers; focuses on the "median" fit.

#### Example

$$\text{minimize } L_\infty = \|Ax - b\|_\infty$$

$$\begin{aligned} \min \quad & t \\ \text{s.t.} \quad & Ax - b \leq t\mathbf{1} \\ & -(Ax - b) \leq t\mathbf{1} \end{aligned}$$

Minimax Approximation or Chebyshev Approximation problem

## 1.3 Forms

Two canonical ways to write an LP. Start from the general form and convert to one of these to hand it to an algorithm.

### 1.3.1 Standard form

All constraints are inequalities, variables non-negative:

$$\begin{aligned} \min_x \quad & c^T x \\ \text{s.t.} \quad & Ax \leq b \\ & x \geq 0 \end{aligned}$$

Conversion rules:

- A  $\geq$  constraint is flipped by multiplying by  $-1$
- An equality  $Fx = g$  becomes two inequalities:  $Fx \leq g$  and  $-Fx \leq -g$
- A free variable  $x_i \in \mathbb{R}$  is split:  $x_i = x_i^+ - x_i^-$  with  $x_i^+, x_i^- \geq 0$

Nice for geometric intuition — the feasible set is a polytope defined by half-spaces.

### 1.3.2 Equality form

Also called *standard equality form* or *slack form*. The **main constraints** are equalities;  $x \geq 0$  is a separate sign restriction (non-negativity bound), not counted among the “constraints” in this naming:

$$\begin{aligned} \min_x \quad & c^T x \\ \text{s.t.} \quad & Ax = b \quad (\text{main constraints — equalities}) \\ & x \geq 0 \quad (\text{sign restriction}) \end{aligned}$$

Conversion rules:

- $Ax \leq b$  becomes  $Ax + s = b$  with slack  $s \geq 0$
- $Ax \geq b$  becomes  $Ax - s = b$  with surplus  $s \geq 0$
- Free variables split the same way as above

This is the form the **simplex algorithm** operates on — it needs equalities so it can pick a basis  $A_B$  (an  $m \times m$  invertible submatrix) and solve  $x_B = (A_B)^{-1}b$  at each vertex. That is why the dimension grows to  $2n + m$ :  $n$  originals +  $n$  for splitting free vars +  $m$  slacks.

### 1.3.3 Quick example

Original problem:

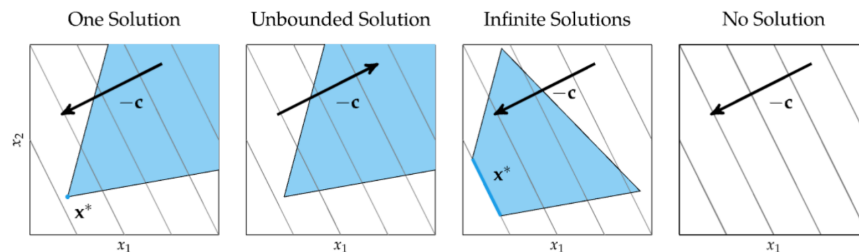
$$\min \quad x_1 + x_2 \quad \text{s.t.} \quad 2x_1 + x_2 \leq 4, \quad x_1, x_2 \geq 0$$

Equality form — add slack  $s \geq 0$ :

$$\min \quad x_1 + x_2 \quad \text{s.t.} \quad 2x_1 + x_2 + s = 4, \quad x_1, x_2, s \geq 0$$

## 2 Simplex Algorithm

Different kinds of problems may lead to different kinds of solutions



## 2.1 Problem Formulation

Linear programs are often solved in **equality form**

$$\begin{aligned} \min_x \quad & c^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0 \\ & x, c \in \mathbb{R}^{2n+m}, \\ & A \in \mathbb{R}^{2n+m} \\ & b \in \mathbb{R}^m \end{aligned}$$

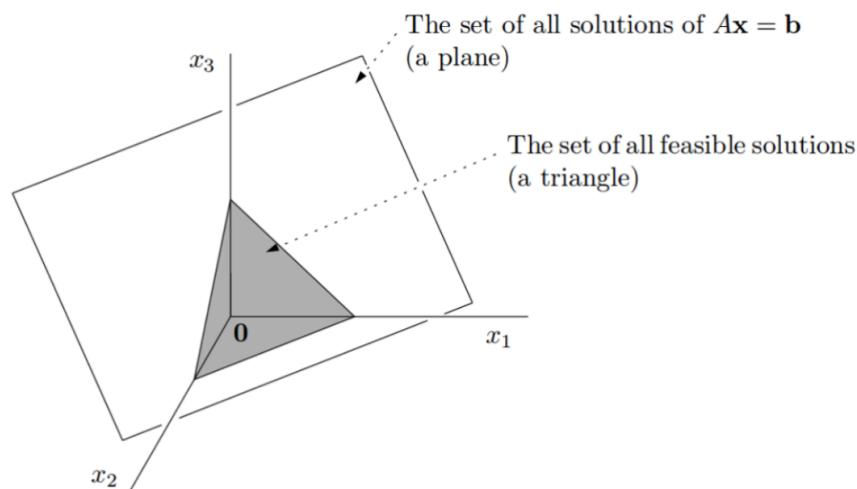


Figure 1: Simplex feasible set

## 2.2 Info abt the simplex algorithm

- Guaranteed to solve any feasible and bounded linear program
- Works on the equality form
- Assumes that rows of  $A$  are linearly independent and  $m \leq n' (n' \leq 2n + m)$
- The feasible set of a linear program forms a polytope. (faces of  $n - 1$  dimension)
- The simplex algorithm moves between vertices of the polytope until it finds an **optimal vertex**

### Theorem (Fundamental Theorem of Linear Programming)

Given:

$$\min \{c^T x \mid x \in P\} \text{ where } P = \{x \in \mathbb{R}^n \mid Ax \leq b\}$$

If  $P$  is a bounded polyhedron and not empty and  $x^*$  is an optimal solution to the problem, then:

- $x^*$  is an extreme point (vertex) of  $P$ , or
- $x^*$  lies on a face  $F \subset P$  of optimal solution



Figure 2: Fundamental Theorem of Linear Programming

We can find these vertices using a system of linear equations.

$$Ax = A_B x_B = b \quad \Leftrightarrow x_B = (A_B)^{-1}b$$

for each vertex. Though this is not really done in practice (very costly).

In practice for this is done by LU-Decomposition [LINK] with

$$\mathcal{O}(3m)$$

Where  $m$  is the number of constraints

## 2.3 The Simplex Algorithm

- every vertex has an associated partition  $(B, N)$
- not every partition corresponds to a vertex.  $A_B$  might be not invertible or the point  $x_B$  might not be  $\geq 0$ .
- identifying partitions that correspond to vertices corresponds to solving an LP problem as well!

The algorithm consists of two phases

- Initialization Phase:** finding a feasible starting vertex
- Optimization Phase:** finding the optimal vertex

### 2.3.1 The problem setup

The LP

$$\begin{aligned} \min_x \quad & c^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0 \\ & x, c \in \mathbb{R}^{2n+m}, \\ & A \in \mathbb{R}^{2n+m} \\ & b \in \mathbb{R}^m \end{aligned}$$

Since  $n > m$ , A vertex is obtained by:  $x_N = 0$

- Picking  $n - m = 2$  variables to set to zero (the non-basis  $N$ )
- Solving for the remaining  $m = 2$  variables (the basis  $B$ )

$$Ax = A_B x_B + A_N x_N = b$$

Since  $x_N = 0$

$$Ax = A_B x_B = b \quad \Rightarrow \quad x_B = A_B^{-1}b$$

Lagrangian function:

$$\mathcal{L}(x, \mu \geq 0, \lambda) = c^T x - \mu^T x - \lambda^T (Ax - b)$$

The necessary conditions for optimality (KKT) are also sufficient for linear programs

### **3 Duality**

### **4 Primal-Dual Hybrid Gradient (PDHG) Method for LP**